Math 55 worksheet Monday 04/27/2026

1. Suppose that we have a bag containing 5 fair dice where 3 of these dice have 6 sides, one has 8 sides, and one has 20 sides. We play a game by selecting 3 dices from our bag uniformly at random, rolling all three and adding their results together. What is the expected value of the sum of our dice?

2. Suppose we are given a fair 8 sided dice and we roll it until a 7 comes up. We may define X to be a random variable representing the number of rolls in such an rolling streak. What is $P(X > 2)$? What is $P(X > 3|X > 1)$?

3. We consider two random processes, in the first we select two numbers x_1 and x_2 uniformly at random from $[1, 2, ..., 100]$ and we define the random variable X to be the product of x_1 with x_2. In the second process we notice that whenever we select two numbers, one must be smaller than or equal to the other, and so we select a number x_1 uniformly at random from $[1, ..., 100]$ and we select x_2 uniformly at random from $[x_1, ..., 100]$ and define the random variable Y to be the product x_1x_2. What are the expected values for these random variables X and Y?

4. Suppose that $X : \Omega \to \mathbb{Z}^{>0}$ is a random variable such that $P(X > s + t|X > s) = P(X > t)$ for all positive integers s and t. Prove that there must exist a $p \in [0, 1]$ such that $P(X = n) = (1 - p)^{n-1}p$ for all positive integers n.